

Firewalld simplifies firewall management by classifying all network traffic into zones. Based on criteria such as the source IP address of a packet or the incoming network's interface, traffic is divided into the firewall rules for the appropriate zone. Each zone can have its own list of ports and services to be opened or closed.

 **Note:** For laptops or other machines that regularly change networks, NetworkManager can be used to automatically set the firewall zone for a connection. The zones can be customised with rules appropriate for particular connections.

This is especially useful when travelling between home, work and public wireless networks. Users might want their system's sshd service to be reachable when connected to their home and corporate networks, but not when connected to the public wireless network in the local coffee shop.

Every packet that comes into the system will be first checked for its source address. If that source address is tied to a specific zone, the rules for that zone will be parsed. If the source address is not tied to a zone, the zone for the incoming network interface will be used.

If the network interface is not associated with a zone for some reason, the default zone will be used. The default zone is not a separate zone in itself but is one of the other zones. The public zone is used by default, but this can be changed by a systems administrator.

Most zones will allow traffic through the firewall that matches a list of particular ports and protocols ('631/udp') or predefined services ('ssh'). If the traffic does not match a permitted port/protocol or service, it will generally be rejected. The trusted zone, which permits all traffic by default, is one exception to this.

Predefined zones

Firewalld ships with a number of predefined zones, suitable for various purposes. The default zone is set to *public* and interfaces are assigned to *public* if no changes are made. The *lo* interface is treated as if it were in a trusted zone. Details of the configuration of these zones on installation follow, but the systems administrator may then customise these zones to have different settings. By default, all zones permit any incoming traffic that is part of a communication initiated by the system, and all outgoing traffic.

Default configuration of firewalld zones

Listed below are the ways in which some of the firewalld zones are configured.

- **Trusted:** Allows all incoming traffic.
- **Home:** Rejects incoming traffic unless related to outgoing traffic or matching the ssh, mdns, ipp-client, samba-client, or dhcpv6-client predefined services.
- **Internal:** Rejects incoming traffic unless related to outgoing traffic or matching the ssh, mdns, ipp-client,

samba-client, or dhcpv6-client predefined services – as in the case of the home zone.

- **Work:** Rejects incoming traffic unless related to outgoing traffic or matching the ssh, ipp-client or dhcpv6-client predefined services.
- **Public:** Rejects incoming traffic unless related to outgoing traffic or matching the ssh or dhcpv6-client predefined services. This is the default zone for newly added network interfaces.
- **External:** Rejects incoming traffic unless related to outgoing traffic or matching the ssh predefined service. Outgoing IPv4 traffic forwarded through this zone masquerades to look like it originated from the IPv4 address of the outgoing network interface.
- **Dmz:** Rejects incoming traffic unless related to outgoing traffic or matching the ssh predefined service.
- **Block:** Rejects all incoming traffic unless related to outgoing traffic.
- **Drop:** Drops all incoming traffic unless related to outgoing traffic (does not even respond with ICMP errors).

For a list of all available predefined zones and their intended uses, you can consult the *firewalld.zones(5)* manual page.

Predefined services

Firewalld also ships with a number of predefined services. These services' definitions can be used to easily permit traffic for a particular network service to pass through the firewall. Given below are details of the predefined services that are used in the default configuration of firewall zones.

- **ssh:** Local SSH server. Traffic to 22/tcp.
- **dhcpv6-client:** Local DHCPv6 client. Traffic to 546/udp on the fe80::/64 IPv6 network.
- **ipp-client:** Local Windows file and print sharing client. Traffic to 137/udp and 138/udp.
- **mdns:** Multi-cast DNS(mDNS) local-link name resolution. Traffic to 5353/udp to the 224.0.0.251(IPv4) or ff02::fb(IPv6) multicast address.

 **Note:** There are many other predefined services. The `firewall-cmd --get-services` command will list them. The configuration files that define the ones included on the firewalld package can be found in the `/usr/lib/firewalld/services` directory, in a format defined by the *firewalld.zone(5)* man page.

The easiest options for systems administrators new to firewalld are to either use predefined services or to explicitly specify the port/protocol they wish to permit. The firewall-config graphical tool can also be used to review predefined services and to define additional services.

Configuring the firewall settings

There are three main ways for systems administrators to interact with firewalld: